STAT 410
Spring 2018

Name ANSWERS

Version A

NetID _____

Exam 2

1	2	3
TuTh	MWF	MWF
2:00 pm	2:00 pm	10:00 am

Please circle your section.

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Total	/ 100

Be sure to show all your work; your partial credit might depend on it.

Put your final answers at the end of your work, and mark them clearly.

No credit will be given without supporting work.

The exam is closed book and closed notes. You are allowed to use a calculator and one 8½" x 11" sheet with notes.

Turn in all scratch paper with your exam.

Academic Integrity

The University statement on your obligation to maintain academic integrity is:

If you engage in an act of academic dishonesty, you become liable to severe disciplinary action. Such acts include cheating; falsification or invention of information or citation in an academic endeavor; helping or attempting to help others commit academic infractions; plagiarism; offering bribes, favors, or threats; academic interference; computer related infractions; and failure to comply with research regulations.

Rule 33 of the Code of Policies and Regulations Applying to All Students gives complete details of rules governing academic integrity for all students. You are responsible for knowing and abiding by these rules.

1. Let $\beta > 0$ and let $X_1, X_2, \ldots, X_n$ be a random sample from the distribution with probability density function

$$f_X(x;\beta) = \frac{\beta^4}{2} x^{11} e^{-\beta x^3}, \qquad x > 0.$$

a) (8) Show that $W = X^3$ has a Gamma distribution. What are its parameters?

No credit will be given without proper justification.

Let $W = X^3$ $\qquad X = W^{1/3} = v(W)$ $\qquad v'(w) = \frac{1}{3} w^{-2/3}$

$$f_W(w) = f_X(v(w)) \cdot |v'(w)| = \frac{\beta^4}{2} w^{11/3} e^{-\beta w} \cdot \frac{1}{3} w^{-2/3} = \frac{\beta^4}{6} w^3 e^{-\beta w}$$

$$= \frac{\beta^4}{\Gamma(4)} w^{4-1} e^{-\beta w}, \qquad w > 0.$$

$\Rightarrow$ W has Gamma$(\alpha = \mathbf{4}, \theta = \mathbf{\frac{1}{\beta}})$ distribution.

b) (6) Suppose $n = 5$ and $\beta = 1.5$. Find $P\left(\sum_{i=1}^{5} X_i^3 > 10\right)$.

$\sum_{i=1}^{n} X_i^3 = \sum_{i=1}^{n} W_i$ has Gamma$(\alpha = 4n, \theta = \frac{1}{\beta})$ distribution.

$$P\left(\sum_{i=1}^{5} X_i^3 > 10\right) = P\left(\text{Gamma}(\alpha = 20, \theta = \frac{1}{1.5}) > 10\right)$$

$$= P\left(\text{Poisson}(1.5 \cdot 10) \le 20 - 1\right) = P\left(\text{Poisson}(15) \le 19\right) = \mathbf{0.875}.$$

1. (continued) $f_X(x;\beta) = \frac{\beta^4}{2} x^{11} e^{-\beta x^3}, \qquad x > 0.$

c) (10) (i) Suggest a $(1-\alpha)100\ \%$ confidence interval for β based on $\sum_{i=1}^{n} X_i^3$.

(ii) Suppose $n = 5, \quad x_1 = 0.3, \quad x_2 = 0.6, \quad x_3 = 0.8, \quad x_4 = 1.1, \quad x_5 = 2.4.$
Construct a 90% confidence interval for β.

$W = X^3$ has $\text{Gamma}(\alpha = 4, \theta = \frac{1}{\beta})$ distribution.

$\Rightarrow \quad \sum_{i=1}^{n} X_i^3$ has $\text{Gamma}(\alpha = 4n, \theta = \frac{1}{\beta})$ distribution.

$\Rightarrow \quad 2\beta \sum_{i=1}^{n} X_i^3$ has a $\chi^2(2\alpha = 8n)$ distribution.

$\Rightarrow \quad P\left(\chi^2_{1-\alpha/2}(8n) < 2\beta \sum_{i=1}^{n} X_i^3 < \chi^2_{\alpha/2}(8n) \right) = 1 - \alpha.$

$\Rightarrow \quad P\left(\frac{\chi^2_{1-\alpha/2}(8n)}{2\sum_{i=1}^{n} X_i^3} < \beta < \frac{\chi^2_{\alpha/2}(8n)}{2\sum_{i=1}^{n} X_i^3} \right) = 1 - \alpha.$

A $(1-\alpha)100\ \%$ confidence interval for β

$$\left(\frac{\chi^2_{1-\alpha/2}(8n)}{2\sum_{i=1}^{n} x_i^3}, \frac{\chi^2_{\alpha/2}(8n)}{2\sum_{i=1}^{n} x_i^3} \right).$$

$\sum_{i=1}^{n} x_i^3 = 15.91.$ $\qquad \chi^2_{0.95}(40) = 26.51, \qquad \chi^2_{0.05}(40) = 55.76.$

$$\left(\frac{26.51}{2 \cdot 15.91}, \frac{55.76}{2 \cdot 15.91} \right) \approx (0.833, 1.752).$$

1. (continued) $f_{\mathrm{X}}(x;\beta) = \dfrac{\beta^4}{2} x^{11} e^{-\beta x^3}, \qquad x > 0.$

d) (4) Find the sufficient statistic $\mathrm{Y} = u(\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_n)$ for β.

$$f(x_1, x_2, \ldots x_n;\beta) = f_{\mathrm{X}}(x_1;\beta) \cdot f_{\mathrm{X}}(x_2;\beta) \cdot \ldots \cdot f_{\mathrm{X}}(x_n;\beta)$$

$$= \left[\frac{\beta^{4n}}{2^n} e^{-\beta \sum_{i=1}^{n} x_i^3} \right] \left(\prod_{i=1}^{n} x_i^{11} \right).$$

By Factorization Theorem, $\mathrm{Y} = \sum_{i=1}^{n} \mathrm{X}_i^3$ is a sufficient statistic for β.

OR

$$f_{\mathrm{X}}(x;\beta) = \exp\left\{ -\beta \cdot x^3 + 4\ln\beta - \ln 2 + 11\ln x \right\}. \qquad \Rightarrow \qquad \mathrm{K}(x) = x^3.$$

$$\Rightarrow \qquad \mathrm{Y} = \sum_{i=1}^{n} \mathrm{K}(\mathrm{X}_i) = \sum_{i=1}^{n} \mathrm{X}_i^3 \text{ is a sufficient statistic for } \beta.$$

1. (continued) $f_X(x;\beta) = \dfrac{\beta^4}{2} x^{11} e^{-\beta x^3}, \qquad x > 0.$

e) (9) (i) Find the maximum likelihood estimator $\hat{\beta}$ of β.

(ii) Suppose $n = 5, \quad x_1 = 0.3, \quad x_2 = 0.6, \quad x_3 = 0.8, \quad x_4 = 1.1, \quad x_5 = 2.4.$
Find the maximum likelihood estimate of β.

$$L(\beta) = \prod_{i=1}^{n}\left(\frac{\beta^4}{2} x_i^{11} e^{-\beta x_i^3} \right) = \frac{\beta^{4n}}{2^n}\left(\prod_{i=1}^{n} x_i^{11} \right) e^{-\beta \sum_{i=1}^{n} x_i^3}$$

$$\ln L(\beta) = 4n \cdot \ln\beta - n \cdot \ln 2 + 11 \sum_{i=1}^{n} \ln x_i - \beta \cdot \sum_{i=1}^{n} x_i^3$$

$$\left(\ln L(\beta)\right)' = \frac{4n}{\beta} - \sum_{i=1}^{n} x_i^3 = 0$$

$$\Rightarrow \qquad \hat{\beta} = \frac{4n}{\sum_{i=1}^{n} X_i^3}.$$

$$\sum_{i=1}^{n} x_i^3 = 0.3^3 + 0.6^3 + 0.8^3 + 1.1^3 + 2.4^3 = 15.91.$$

$$\hat{\beta} = \frac{20}{15.91} \approx 1.257.$$

1. (continued) $f_X(x;\beta) = \frac{\beta^4}{2} x^{11} e^{-\beta x^3}, \qquad x > 0.$

f) (9) Is the maximum likelihood estimator $\hat{\beta}$ an unbiased estimator of β?
If $\hat{\beta}$ is not an unbiased estimator of β, construct an unbiased estimator of β based on $\hat{\beta}$.

$Y = \sum_{i=1}^{n} X_i^3 = \sum_{i=1}^{n} W_i$ has Gamma$(\alpha = 4n,\ \theta = \frac{1}{\beta})$ distribution.

$$\hat{\beta} = \frac{4n}{\sum_{i=1}^{n} X_i^3} = \frac{4n}{Y}.$$

$T_\alpha \sim \text{Gamma}(\alpha,\ \theta = \frac{1}{\lambda}) \qquad \Rightarrow \qquad E(T_\alpha^k) = \frac{\theta^k \Gamma(\alpha+k)}{\Gamma(\alpha)} = \frac{\Gamma(\alpha+k)}{\lambda^k \Gamma(\alpha)}.$

$$E(\hat{\beta}) = E\left(\frac{4n}{Y}\right) = 4n\, E\left(\frac{1}{Y}\right) = 4n\, E(Y^{-1}) = 4n \cdot \frac{\beta}{4n-1} = \frac{4n}{4n-1} \cdot \beta \neq \beta.$$

$\Rightarrow$ $\hat{\beta}$ is NOT an unbiased estimator for β.

Consider $\hat{\hat{\beta}} = \frac{4n-1}{4n} \cdot \hat{\beta} = \frac{4n-1}{\sum_{i=1}^{n} X_i^3}.$

Then $E(\hat{\hat{\beta}}) = \frac{4n-1}{4n} \cdot E(\hat{\beta}) = \beta.$ $\hat{\hat{\beta}}$ is an unbiased estimator for β.

2. Let $a > 0$ and let $X_1, X_2, \ldots, X_n$ be a random sample from the distribution with probability density function

$$f(x; a) = \frac{4\,a^4}{x^5}, \qquad x \geq a.$$

a) (9) (i) Find a method of moments estimator $\tilde{a}$ of a.

(ii) Suppose $n = 4$, and $x_1 = 2.3,\ x_2 = 2.5,\ x_3 = 3.1,\ x_4 = 4.1.$
Find a method of moments estimate of a.

$a = \theta, \qquad \theta = a.$

$$E(X) = \int_{\theta}^{\infty} x \cdot \frac{4\,\theta^4}{x^5}\,dx = \frac{4}{3}\,\theta.$$

$$\overline{X} = \frac{4}{3}\,\tilde{\theta}. \qquad \Rightarrow \qquad \tilde{\theta} = \frac{3}{4}\,\overline{X}.$$

$x_1 = 2.3,\ x_2 = 2.5,\ x_3 = 3.1,\ x_4 = 4.1.$ $\qquad \overline{x} = 3.$

$\tilde{\theta} = \mathbf{2.25}.$

2. (continued) $f(x;a) = \dfrac{4\,a^4}{x^5}, \qquad x \geq a.$

b) (9) (i) Find the maximum likelihood estimator $\hat{a}$ of a.

(ii) Suppose $n = 4$, and $x_1 = 2.3,\ x_2 = 2.5,\ x_3 = 3.1,\ x_4 = 4.1.$

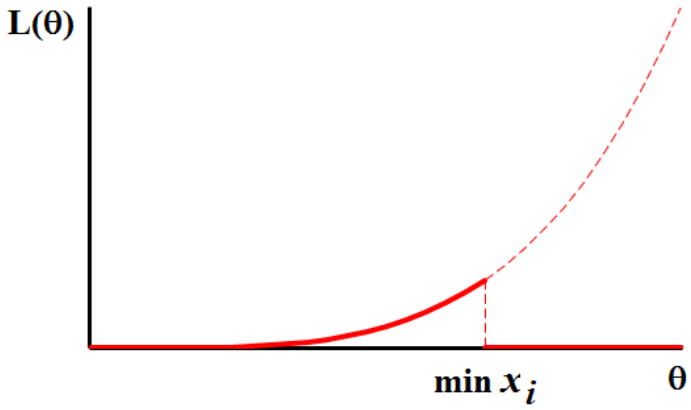
Find the maximum likelihood estimate of a.

$a = \theta, \qquad \theta = a.$

$$L(\theta) = \prod_{i=1}^{n} \frac{4\,\theta^4}{x_i^5} = 4^n\,\theta^{4n} \prod_{i=1}^{n} x_i^{-5}, \qquad x_1 \geq \theta,\ x_2 \geq \theta,\ \ldots,\ x_n \geq \theta,$$

$= 0,$ otherwise.

$x_1 \geq \theta,\ x_2 \geq \theta,\ \ldots,\ x_n \geq \theta \quad \Leftrightarrow \quad \theta \leq \min x_i.$



$\Rightarrow \quad \hat{\theta} = \mathbf{\min X_i}.$

$x_1 = 2.3,\ x_2 = 2.5,\ x_3 = 3.1,\ x_4 = 4.1. \qquad \hat{\theta} = \min x_i = \mathbf{2.3}.$

2. (continued) $f(x;a) = \dfrac{4a^4}{x^5}, \qquad x \geq a.$

c) (9) Is the maximum likelihood estimator $\hat{a}$ an unbiased estimator of a? If $\hat{a}$ is not an unbiased estimator of a, construct an unbiased estimator of a based on $\hat{a}$.

$a = \theta, \qquad \theta = a.$

$$F_X(x) = \int_\theta^x \frac{4\theta^4}{u^5}\,du = 1 - \frac{\theta^4}{x^4}, \qquad x \geq \theta.$$

$$F_{\min X_i}(x) = 1 - (1 - F_X(x))^n = 1 - \frac{\theta^{4n}}{x^{4n}}, \qquad x \geq \theta.$$

$$f_{\min X_i}(x) = \frac{4n\theta^{4n}}{x^{4n+1}}, \qquad x \geq \theta.$$

$$E(\hat{\theta}) = E(\min X_i) = \int_\theta^\infty x \cdot \frac{4n\theta^{4n}}{x^{4n+1}}\,dx = \frac{4n\theta}{4n-1} \neq \theta.$$

$\Rightarrow$ $\hat{\theta}$ is NOT an unbiased estimator for θ.

Consider $\hat{\hat{\theta}} = \dfrac{4n-1}{4n} \cdot \hat{\theta} = \dfrac{4n-1}{4n} \cdot \min X_i.$

Then $E(\hat{\hat{\theta}}) = \dfrac{4n-1}{4n} \cdot E(\hat{\theta}) = \theta.$ $\hat{\hat{\theta}}$ is an unbiased estimator for θ.

3. Ex-Coin and Why-Coin are two (much) lesser known cyber currencies; their prices \$X and \$Y (respectively) vary from day to day according to a bivariate normal distribution with parameters

$$\mu_X = 72, \quad \sigma_X = 7, \quad \mu_Y = 57, \quad \sigma_Y = 5, \quad \rho = 0.8.$$

a) (9) Suppose that on a given day the price of Why-Coin is \$61. What is the probability that the price of Ex-Coin exceeds \$79? That is, find $P(X > 79 \mid Y = 61)$.

Given $Y = 61$, X has Normal distribution

with mean $\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y) = 72 + 0.8 \cdot \frac{7}{5} \cdot (61 - 57) = 76.48$

and variance $\left(1 - \rho^2\right) \cdot \sigma_X^2 = \left(1 - 0.8^2\right) \cdot 7^2 = 17.64$

(standard deviation $= 4.2$).

$$P(X > 79 \mid Y = 61) = P\left(Z > \frac{79 - 76.48}{4.2}\right) = P(Z > 0.60) = \mathbf{0.2743}.$$

b) (9) Alex buys 4 Ex-Coins and 3 Why-Coins. What is the probability that the value of this portfolio exceeds \$500? That is, find $P(4X + 3Y > 500)$.

$4X + 3Y$ has Normal distribution,

$$E(4X + 3Y) = 4\mu_X + 3\mu_Y = 4 \cdot 72 + 3 \cdot 57 = 459,$$

$$\begin{aligned} \mathrm{Var}(4X + 3Y) &= 16\,\sigma_X^2 + 24\,\sigma_{XY} + 9\,\sigma_Y^2 \\ &= 16\,\sigma_X^2 + 24\,\rho\,\sigma_X\sigma_Y + 9\,\sigma_Y^2 \\ &= 16 \cdot 7^2 + 24 \cdot 0.8 \cdot 7 \cdot 5 + 9 \cdot 5^2 = 1{,}681 \end{aligned}$$

(standard deviation $= 41$).

$$P(4X + 3Y > 500) = P\left(Z > \frac{500 - 459}{41}\right) = P(Z > 1.00) = \mathbf{0.1587}.$$

4. (9) Let $a > 0$, and let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a distribution with probability density function

$$f_X(x; a) = a \cdot x^{a-1}, \qquad 0 < x < 1.$$

Construct a consistent estimator of a based on $\sum_{i=1}^{n} X_i^5$.

$a = \theta, \qquad \theta = a.$

By WLLN, $\overline{X^5} = \frac{1}{n} \cdot \sum_{i=1}^{n} X_i^5 \xrightarrow{P} E(X^5) = \int_0^1 x^5 \cdot \theta\, x^{\theta-1}\, dx = \frac{\theta}{\theta+5}.$

$X_n \xrightarrow{P} a$, g is continuous at $a \Rightarrow g(X_n) \xrightarrow{P} g(a)$

Consider $g(x) = \frac{5x}{1-x}$. Since $g(x)$ is continuous at $\frac{\theta}{\theta+5}$,

$$\frac{5\sum_{i=1}^{n} X_i^5}{n - \sum_{i=1}^{n} X_i^5} = \frac{5\overline{X^5}}{1-\overline{X^5}} = g(\overline{X^5}) \xrightarrow{P} g\left(\frac{\theta}{\theta+5}\right) = \theta.$$